The 2023-09-08 Al Haouz, Morocco earthquake (M_w 6.8, us7000kufc): a reproducible MTUQ + WISP validation study of a shallow continental thrust

M.-S. Seo

MTUQ (point-source moment tensor) + WISP (kinematic finite fault); every parameter documented

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Abstract

We determine the point-source moment tensor (MTUQ) and a kinematic finite-fault model (WISP) of the 2023-09-08 22:11:01 UTC M_w 6.8 Al Haouz earthquake — a shallow (∼26 km) oblique-reverse thrust in the Moroccan High Atlas that killed ∼2,900 people. Unlike the mostly-unstudied events we analysed previously, this one has a USGS moment tensor and finite-fault model, so we use it to validate the pipeline and to document every data, frequency-band and processing parameter for reproducibility. Our double-couple moment tensor, 256/67/65 (M_w 6.95), lies 5.1° (Kagan angle) from the USGS W-phase solution; our WISP finite fault on the steep plane (peak slip 1.46 m at 17–37 km, ∼43×38 km) closely reproduces the USGS model (1.88 m, 25 km, 45×30 km) and prefers the same steep nodal plane. The study also yields a methodological result that mirrors our deep-event work: for a shallow source the joint body-plus-surface-wave inversion returns a confidently wrong (flipped, normal-faulting) mechanism, because surface waves are dip-slip-indeterminate at shallow depth (Kanamori & Given, 1981) and mismatched by 1-D dispersion; the body waves, through their depth phases, carry the correct dip-slip sense. Inverting body-only recovers the USGS mechanism.

Plain-language summary

A magnitude-6.8 earthquake struck the High Atlas mountains south of Marrakesh, on a reverse (compressional) fault about 26 km deep. Because this event is already well studied, we use it to check our analysis tools against the U.S. Geological Survey's published answers — and they agree closely. Along the way we found a trap specific to shallow earthquakes: the long-period "surface waves" cannot tell which way such a fault slipped and gave the opposite (pull-apart) answer; the shorter-period "body waves" gave the correct (push-together) answer. This is the mirror image of what we learned for very deep earthquakes, where the body waves were the unreliable ones.

1 Event and USGS benchmarks

The earthquake (31.058°N, 8.385°W) is an intra-continental oblique-reverse rupture in the High Atlas. The USGS W-phase moment tensor is 122/29/132 (shallow SE-dipping) with conjugate 255/69/69 (steep NW-dipping), 94% double couple, centroid 30.5 km; the Mwc (26 km) and Mwb (24 km) solutions and the 19-km catalogue hypocentre span a genuine depth debate. The USGS finite-fault model adopted the steep plane (strike 251°, dip 69°), 45×30 km, peak slip 1.88 m, depth 25 km. These are the numbers we reproduce.

2 Data and parameters (documented for reproducibility)

Determinism. FDSN data fetching is deterministic in its parameters but not across time (data-centre availability changes; transient request failures drop stations). The reproducible unit is therefore the archived waveforms plus the parameters below: re-running from the saved SAC files is byte-deterministic, while re-fetching yields a near-identical set. The inversions are deterministic (MTUQ is an exhaustive grid search; WISP's annealing reproduces to four decimals). Compute is CPU-only (no GPU): MTUQ parallelises with MPI (`mpirun -n 20`); WISP uses OpenMP (Green's functions and annealing) plus a `multiprocessing` pool for data processing.

item	value
WISP data	`ffm manage acquire -t body` (II, G, IU, GE), ∼150 broadband stations
MTUQ data	20 GSN broadband, 31–86°, reused from the WISP fetch; response via StationXML inventory (output DISP), rotated ZNE→RT
MTUQ body band	15–40 s (`--bwband 15,40,80,8`); 80-s window, ±8 s shifts
MTUQ surface band	50–100 s (shown to flip the shallow mechanism; excluded from the solution)
MTUQ Green's functions	syngine ak135 (`ak135f_2s`)
MTUQ grid	DoubleCoupleGridRegular, 40 pts/axis (∼64k orientations × 5 magnitudes)
WISP body filter	0.006–1.0 Hz, wavelet scales 2–8
WISP surface waves	used (shallow source; above the 125-km surface-wave GF cap)
adopted depth	26 km

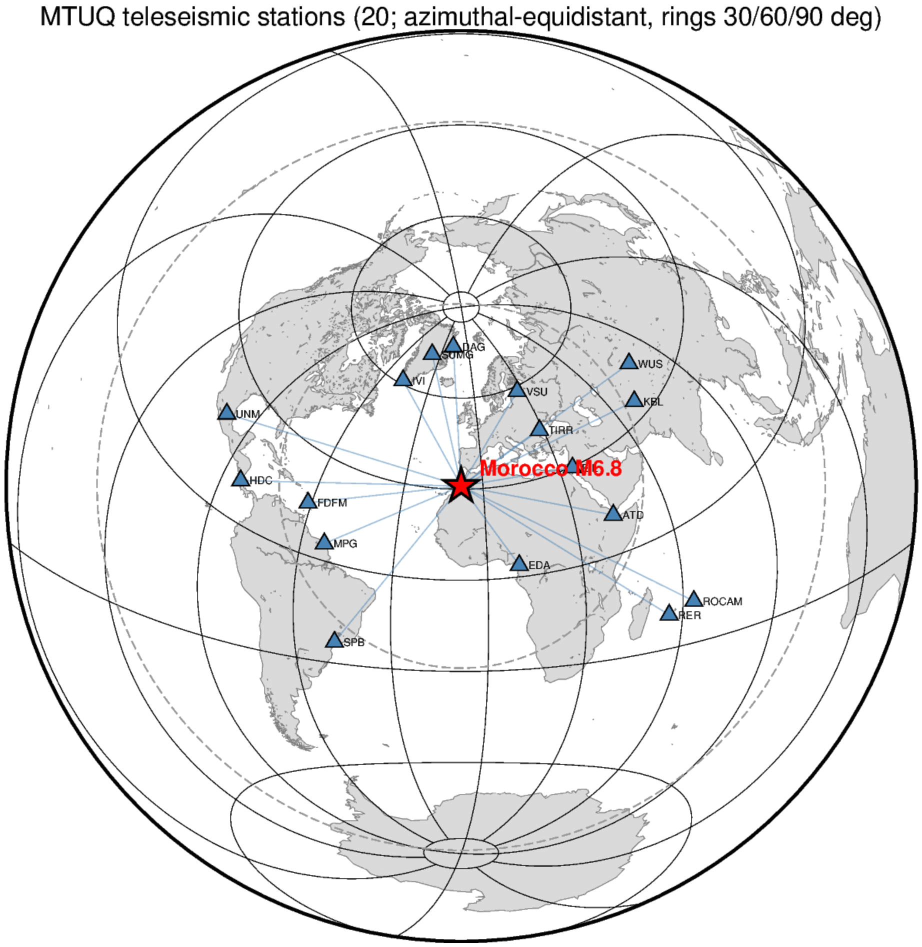

Figure 1: The 20 GSN teleseismic broadband stations used for the moment tensor, on an azimuthal-equidistant projection centred on the epicentre (rings at 30/60/90°).

3 Moment tensor: the shallow-event trap and cure

Run jointly (body 10–33 s + surface 50–100 s), the DC search converges confidently on a flipped, normal-faulting mechanism ~89° (Kagan) from USGS. The cause is physical: at shallow depth the vertical dip-slip moment-tensor elements (M_{rt}, M_{rp}) are weakly excited at long period — the Kanamori–Given shallow indeterminacy — and the surface waves are additionally mismatched by 1-D dispersion (large time shifts, poor amplitude fit), so they cannot fix, and actively mislead, the dip-slip sense. The body waves, through their depth phases, resolve it. We therefore invert body-only (a `--onlybody` option added to the runners). This is the mirror image of the deep-event rule, where short-period body waves fail and the surface/long-period waves are reliable.

solution	M_w	strike/dip/rake	Kagan vs USGS Mww	%DC
DC (adopted)	6.95	256/67/65	5.1°	—
full moment tensor	6.80	270/63/63	19.1°	66% (USGS 94%)

Table 1: Body-only MTUQ. The DC solution matches USGS to 5.1°; the full MT is destabilised by the same shallow-source indeterminacy (spurious non-DC, 66% vs 94%), which is exactly why the DC constraint is standard for shallow sources. We adopt the DC solution.

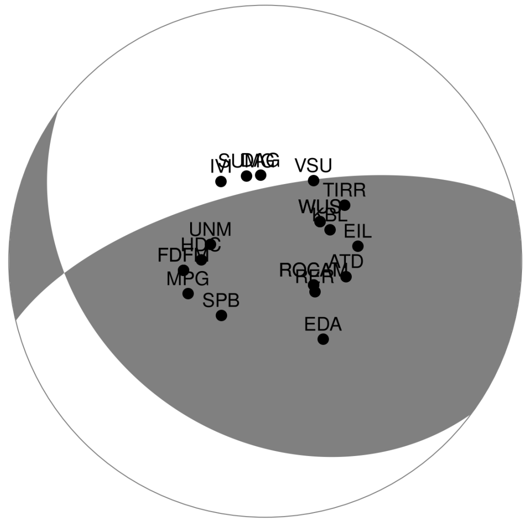

Figure 2: Adopted DC moment tensor (256/67/65) with the teleseismic station distribution — the correct oblique-reverse thrust.

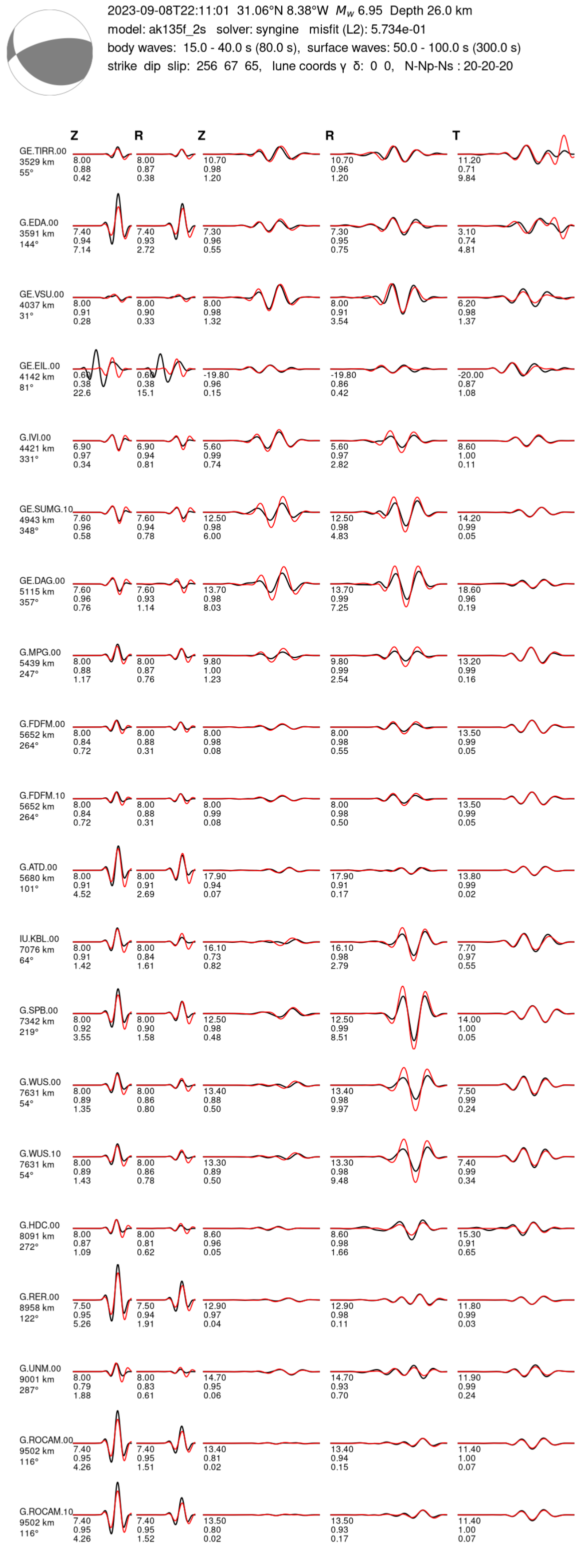

Figure 3: Body-wave fits of the adopted DC solution (black observed, red synthetic).

4 Finite fault: validation against the USGS model

We ran WISP `auto_model` on both nodal planes of our own MTUQ mechanism (internal consistency: the fault planes come from our moment tensor, not an external CMT). Being shallow, surface waves are fully used.

plane	misfit	peak slip	slip depth (>15%)	$L \times W$	USGS finite fault
steep 256/67 (preferred)	0.1364	1.46 m	17–37 km	43×38 km	251/69, 1.88 m, 25 km, 45×30 km
shallow 127/25	0.1402	1.50 m	11–44 km	39×38 km	—

Table 2: The steep plane is preferred (agreeing with the USGS choice), and its slip model closely reproduces the USGS finite fault in amplitude, depth and dimension. As for the Chile events the teleseismic fit difference between planes is small; the steep-plane preference is corroborated by USGS aftershocks and InSAR.

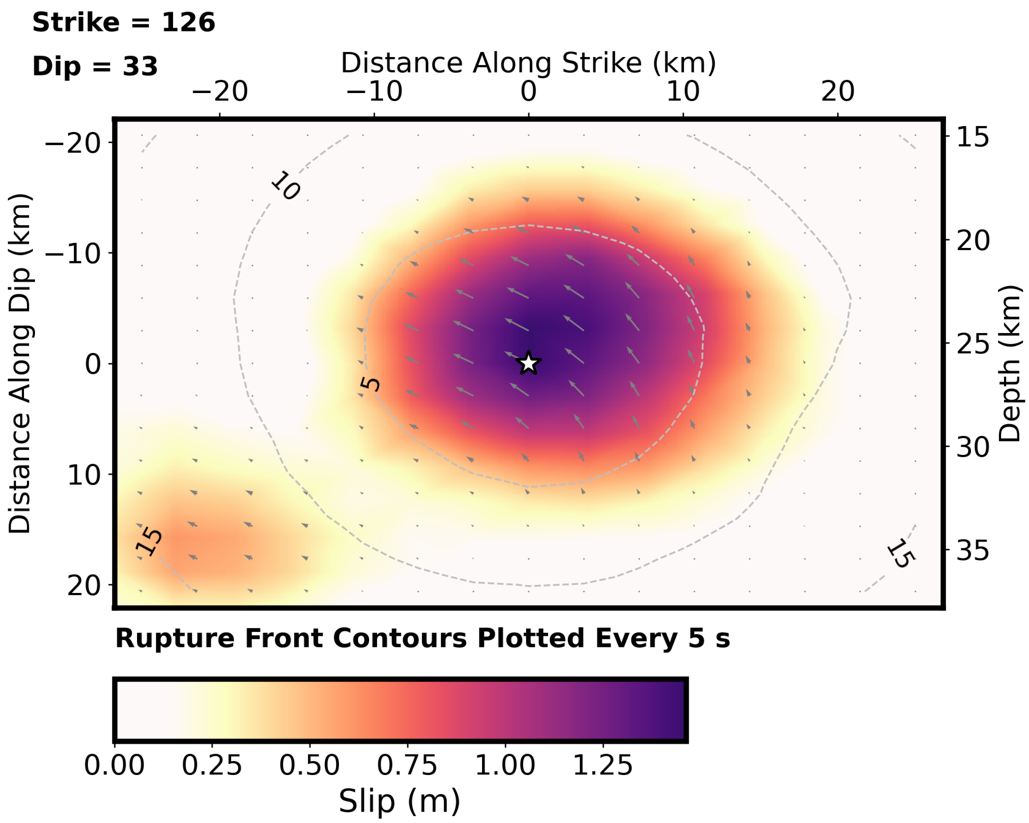

Figure 4: Preferred (steep-plane) slip model. Star: hypocentre; contours: rupture front every 5 s; arrows: slip direction (reverse). Peak 1.46 m, slip at 17–37 km depth.

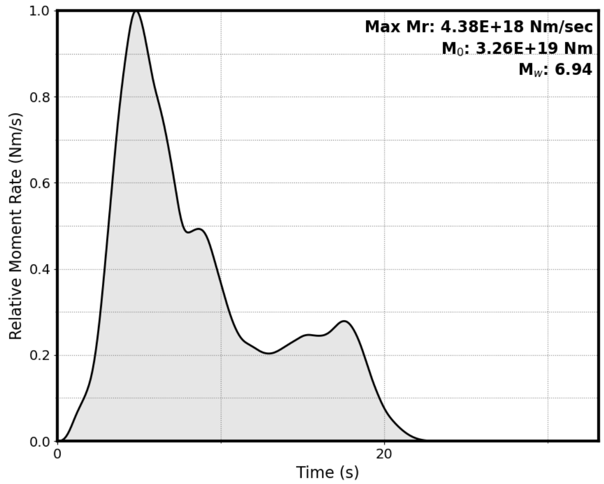

Figure 5: Moment-rate function (steep plane).

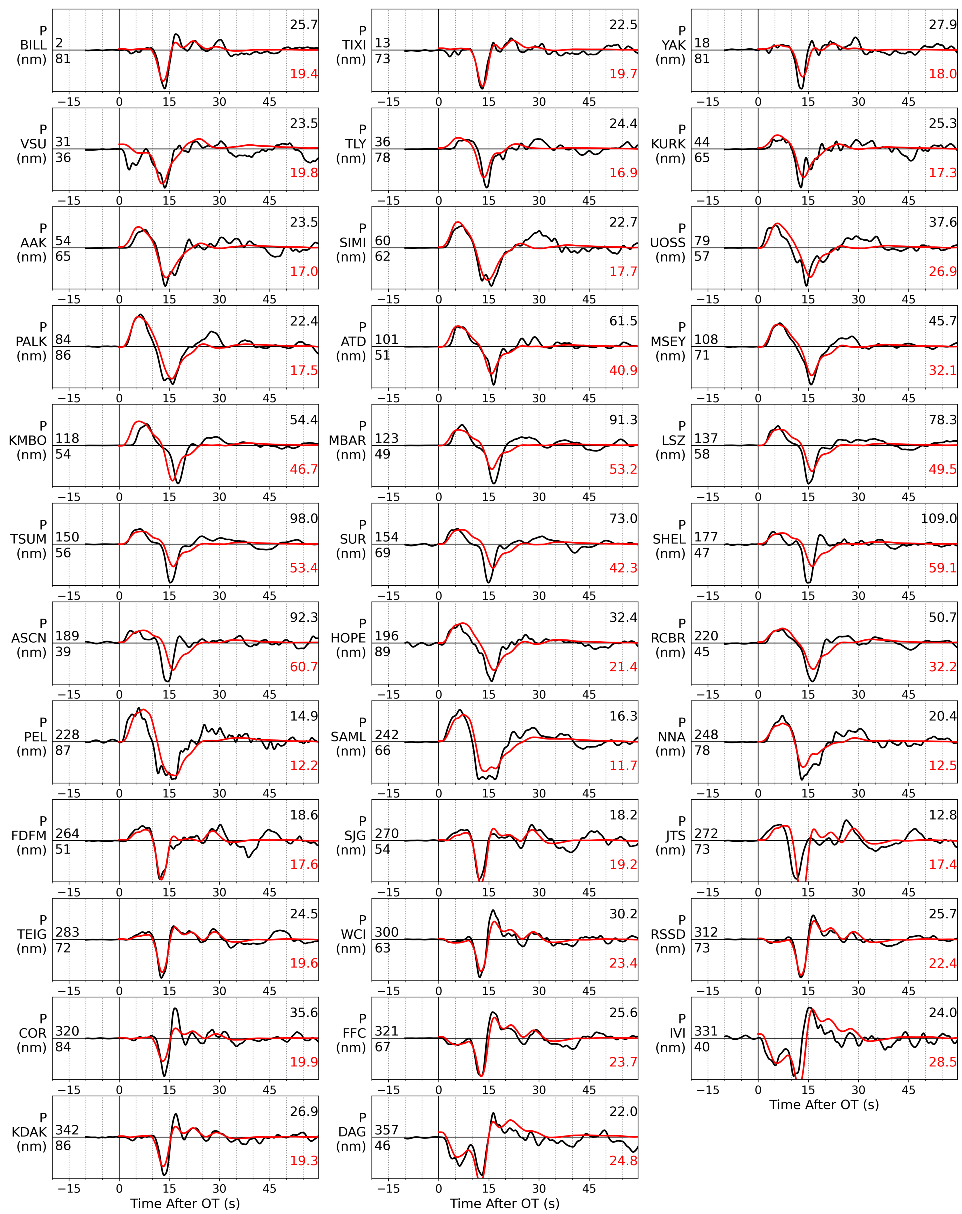

Figure 6: WISP P-wave fits (steep plane).

Figure 7: WISP Rayleigh-wave fits (steep plane).

5 Summary

1. The pipeline reproduces the USGS moment tensor to 5.1° and the USGS finite-fault model (slip amplitude, depth, dimensions, preferred plane) for this well-studied event — a clean end-to-end validation across a new tectonic regime (shallow continental thrust).

2. Shallow-event rule: invert body waves for the mechanism; surface waves are dip-slip-indeterminate at shallow depth and flip the solution. Mirror of the deep-event rule.

3. Use the DC constraint for shallow sources — the full MT is destabilised by the same indeterminacy (66% vs USGS 94% DC).

4. Every parameter is tabulated; the analysis re-runs deterministically from the archived waveforms. Pipeline is CPU-parallel (MPI + OpenMP), no GPU.